Finite Excess in Ordinal Nim Multiplication

Exact Order Reductions and the Open 0/1/4 Rule

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Abstract

The finite part of a Kummer carry in Conway's field of ordinal nimbers is Lenstra's excess m_p, indexed by an odd prime p. Every presently certified value belongs to $\{0, 1, 4\}$, with the value predicted by the component structure of $\mathrm{ord}_p(2)$. We give an exact reduction of this rule to four universal multiplicative-order assertions. They concern, respectively, a synchronized structural norm, a selected projective class on every ordinary odd Kummer spine, a Gaussian period in a cubic norm tower, and a corrected norm on the exceptional $2 \cdot 3^k$ spine. The reduction retains the full primary valuation in each finite cyclic group; testing only divisibility of an order by p is not sufficient in general.

For each arm we record a concise selected-coordinate normal form. The singleton-even part becomes a Conway–Fermat quotient-order problem and a zero-at-multiples Fibonacci recurrence. The ordinary arm becomes a power-residue symbol of one marked cyclotomic unit. The cubic arm becomes generation by one recursively selected element of a norm-one torus. The exceptional arm becomes irreducibility of an explicit Capelli composition after a corrected quadratic norm. In that exceptional column we also obtain the unconditional level-dependent bound $m_\ell < 16(3^k)^4$ when $f(\ell) = 2 \cdot 3^k$ and $k \geq 2$. We further identify the literal Conway–Fermat points on a fixed supersingular elliptic curve, prove that the ordinary marked primes form a unique inert cyclotomic ray, and show that the cubic recursion and the exceptional auxiliary ancestry are affine forms of one toric Dickson system. These structures sharpen the selected phase which remains to be evaluated; they do not force its nonvanishing. These are reductions, not proofs of the four universal assertions. Formal verification establishes the general group, norm, and polynomial identities and several finite arithmetic certificates; its universal exceptional target is a definition rather than a theorem. Finite computations are reported separately and are not used as evidence for a universal conclusion.

1 Introduction

Conway's nim addition and nim multiplication make the ordinals into a proper class field of characteristic two. Its finite subfields are canonically embedded, and its initial transfinite part is assembled by a sequence of Kummer extensions [3, 7, 8]. Lenstra's description isolates, for each odd prime p, a carry

$$\alpha_p = \kappa_{f(p)} + m_p, \qquad f(p) = \mathrm{ord}_p(2),$$

where $\kappa_{f(p)}$ is a structural tower element and the natural number m_p is the *finite excess*. The same construction is part of the general structural description of ordinal fields On_p [5].

Let $\mathcal{Q}(h)$ denote Lenstra's component set for κ_h. The available data suggest the following rigid rule.

Conjecture 1.1 (the 0/1/4 rule)**.** *For every odd prime p,*

$$m_p = \begin{cases} 0, & \mathcal{Q}(f(p)) \text{ is not a singleton odd prime power}, \\ 4, & f(p) = 2 \cdot 3^k \text{ for some } k \geq 1, \\ 1, & \text{otherwise.} \end{cases}$$

The purpose of this article is to state the exact mathematical frontier of Conjecture 1.1. Theorem 3.1 shows that it is equivalent to four selected order assertions, denoted Z, O, C, D. None of them is known in full. The word *selected* is essential: each field contains many elements with the same degree, trace, norm, or generic cyclotomic shape, while the nim construction chooses one recursively marked Frobenius orbit. The open content is the multiplicative order of that orbit.

The main results are organized as follows. Section 2 gives the exact power and norm criteria. Section 3 states and proves the four-arm equivalence. Sections 4–7 give one lossless normal form for each arm. Section 8 separates the stronger 0/1/4 rule from Lenstra's boundedness question. Section 9 states the formal and computational boundary.

2 Finite-field and tower preliminaries

For a finite algebraic nimber x, write

$$d(x) = [\mathbb{F}_2(x) : \mathbb{F}_2].$$

The element κ_h and its component set $\mathcal{Q}(h)$ are those of Lenstra's tower decomposition. In particular, κ_h is the sum of the selected components κ_q, $q \in \mathcal{Q}(h)$, and $h \mid d(\kappa_h)$. The excess is equivalently

$$m_p = \min\left\{m \geq 0 : \kappa_{f(p)} + m \text{ is not a } p\text{-th power in its defining finite field}\right\}.$$

Finite integers in this formula are identified with their canonical finite nimbers. We use standard finite-field facts, including cyclicity of the multiplicative group, the norm formula, and Capelli's irreducibility lemma, as collected in [9].

Proposition 2.1 (exact power criterion)**.** *Let p be a prime, let $p \mid 2^E - 1$, and let $x \in \mathbb{F}_{2^E}^\times$. Then the following are equivalent:*

1. *x is a p-th power in $\mathbb{F}_{2^E}^\times$;*
2. *$x^{(2^E-1)/p} = 1$;*
3. *$\operatorname{ord}(x) \mid (2^E - 1)/p$.*

Consequently, x is not a p-th power if and only if

$$v_p(\operatorname{ord}(x)) = v_p(2^E - 1).$$

Proof. Choose a generator g of the cyclic group of order $2^E - 1$ and write $x = g^a$. The congruence $py \equiv a \pmod{2^E - 1}$ is soluble exactly when $p \mid a$, which is equivalent to $x^{(2^E-1)/p} = 1$. The order formulation follows from $x^n = 1$ if and only if $\operatorname{ord}(x) \mid n$. The final statement retains the full p-primary valuation. □

Remark 2.2 (the valuation cannot be discarded). If $h = f(p)$ and $h \mid E$, lifting the exponent gives

$$v_p(2^E - 1) = v_p(2^h - 1) + v_p(E/h).$$

Thus the shortcut $p \mid \operatorname{ord}(x)$ is equivalent to nonexistence of a p-th root only when the ambient p-part has valuation one. Every statement below uses Proposition 2.1, not the shortcut.

Proposition 2.3 (norm reduction). *Let $h \mid E$, let $p \mid 2^h - 1$, and let $x \in \mathbb{F}_{2^E}^\times$. Then*

$$x^{(2^E-1)/p} = \mathrm{N}_{\mathbb{F}_{2^E}/\mathbb{F}_{2^h}}(x)^{(2^h-1)/p}.$$

In particular, x is a p-th power in $\mathbb{F}_{2^E}$ if and only if its norm is a p-th power in $\mathbb{F}_{2^h}$.

Proof. The norm is $x^{(2^E-1)/(2^h-1)}$. Raising it to $(2^h - 1)/p$ gives the displayed identity. Proposition 2.1 gives the last assertion. □

The last equivalence is also visible on Kummer quotients: extension and norm multiply the class by the relative degree. In the applications below, the relative degrees are prime to the current prime, so no selected primary coordinate is lost.

3 The four exact arms

We first define the four order assertions. For $h > 1$, put

$$\Pi_h = \prod_{\substack{\ell \text{ odd prime} \\ \operatorname{ord}_\ell(2)=h}} \ell^{v_\ell(2^h-1)}, \qquad T_h = \mathrm{N}_{\mathbb{F}_{2^{d(\kappa_h)}}/\mathbb{F}_{2^h}}(\kappa_h).$$

The product Π_h is the complete primary part of $\mathbb{F}_{2^h}^\times$ supported at primes for which h is the first residue degree.

For an odd prime r, let $b_r = d(\alpha_r)$, and for $a \geq 1$ put

$$x_{r,a} = \kappa_{r^a}, \qquad F_{r,a} = \mathbb{F}_{2^{r^a b_r}}, \qquad y_{r,a} = x_{r,a} + 1,$$

with $F_{r,0} = \mathbb{F}_{2^{b_r}}$. Define the cyclic projective quotient

$$G_{r,a} = F_{r,a}^\times / F_{r,a-1}^\times, \qquad L_{r,a} = |G_{r,a}| = \Phi_{r^a}(2^{b_r}).$$

For $k \geq 1$, set $h = 3^k$, let $\zeta_k = \kappa_{3^k}$, and define

$$\gamma_k = \zeta_k + \zeta_k^{-1} \in \mathbb{F}_{2^h}^\times.$$

Finally, put $q = 2^h$, $\omega = \zeta_k^h \in \mathbb{F}_4$, and

$$A_k = \zeta_k^2 + \zeta_k + \omega, \qquad M_k = A_k^{q-1}, \qquad \Psi_k = \frac{\Phi_{2\cdot 3^k}(2)}{3}.$$

Theorem 3.1 (four-arm reduction). *Conjecture 1.1 is equivalent to the conjunction of the following four universal assertions.*

Z. For every $h > 1$ such that $\mathcal{Q}(h)$ is not a singleton odd prime power,

$$\Pi_h \mid \operatorname{ord}(T_h).$$

O. For every odd prime $r \neq 3$, every $a \geq 1$, and every prime $p \neq r$ dividing $L_{r,a}$,

$$v_p\Big(\operatorname{ord}(y_{r,a}F_{r,a-1}^{\times})\Big) = v_p(L_{r,a}).$$

C. The element γ_k is primitive in $\mathbb{F}_{2^{3^k}}^{\times}$ for every $k \geq 1$.

D. One has

$$\Psi_k \mid \operatorname{ord}(M_k) \qquad (k \geq 1).$$

Proof. Fix an odd prime p and write $h = f(p)$. If $\mathcal{Q}(h)$ is not a singleton odd prime power, the proposed value is zero. Proposition 4.1 below identifies the simultaneous zero claim at residue degree h with Z.

Suppose next that $\mathcal{Q}(h) = \{r^a\}$, where r is odd. Lenstra's degree and component formulas [7, 8] give

$$h = r^a s, \qquad s \mid b_r, \qquad \kappa_h = x_{r,a}.$$

Proposition 5.1 shows that the row primes are exactly the prime divisors $p \neq r$ of $L_{r,a}$, and that $m_p = 1$ is equivalent to the full primary-order equality in O.

It remains to separate $r = 3$. Here $b_3 = 2$, so $h = 3^a$ or $h = 2 \cdot 3^a$. Proposition 6.1 identifies the first case with primitivity of γ_a, which is C. Propositions 7.1 and 7.3 identify the second case with D. The cases are disjoint and exhaustive. $\square$

Remark 3.2 (status of the theorem). Theorem 3.1 is an equivalence. It does not establish any of the four universal assertions. The known finite cases and the exact formalization boundary are stated in Section 9.

4 The zero arm

4.1 Structural norm and synchronized phase

Proposition 4.1 (exact support criterion). *Let $h > 1$, let $E = d(\kappa_h)$, and let $T_h = \mathrm{N}_{\mathbb{F}_{2^E}/\mathbb{F}_{2^h}}(\kappa_h)$. For an odd prime p with $f(p) = h$,*

$$m_p = 0 \iff v_p(\operatorname{ord}(T_h)) = v_p(2^h - 1).$$

Consequently, $m_p = 0$ for every such p if and only if $\Pi_h \mid \operatorname{ord}(T_h)$.

Proof. By definition, $m_p = 0$ exactly when κ_h is not a p-th power in $\mathbb{F}_{2^E}$. Proposition 2.3 moves the test to $T_h \in \mathbb{F}_{2^h}$, and Proposition 2.1 gives the valuation equality. Multiplying the independent primary conditions gives the final assertion. $\square$

When κ_h has several components, the structural norm is not the product of separate component norms. The missing datum is a synchronized Frobenius phase, made explicit by the following general construction.

Proposition 4.2 (synchronized additive resultant). *Let $K = \mathbb{F}_Q$, let $p \mid Q - 1$, and choose $u_i \in L_i = \mathbb{F}_{Q^{r_i}}$ of degree r_i over K. Put*

$$R = \operatorname{lcm}(r_1, \ldots, r_t), \qquad M = \mathbb{F}_{Q^R}, \qquad u = \sum_{i=1}^{t} u_i,$$

and assume $u \neq 0$ *and* $p \nmid R$. *Let*

$$\Omega = \left(\prod_i \mathbb{Z}/r_i\mathbb{Z} \right) / \langle (1, \ldots, 1) \rangle.$$

For a phase $\mathbf{d} = (d_i)$, *define*

$$F_{\mathbf{d}}(Z) = \prod_{j=0}^{R-1} \left(Z + \sum_i u_i^{Q^{j+d_i}} \right).$$

Then $F_{\mathbf{d}} \in K[Z]$, *it depends only on the class of* $\mathbf{d}$, *and*

$$\prod_{0 \le j_i < r_i} \left(Z + \sum_i u_i^{Q^{j_i}} \right) = \prod_{\mathbf{d} \in \Omega} F_{\mathbf{d}}(Z).$$

The selected common embedding is the zero phase, for which

$$F_{\mathbf{0}}(0) = \mathrm{N}_{M/K}(u), \qquad [F_{\mathbf{0}}(0)] = R[u] \quad \text{in } M^\times / M^{\times p}.$$

Hence u *is a non-p-th-power if and only if the selected constant term* $F_{\mathbf{0}}(0)$ *is a non-p-th-power. Separate component minimal polynomials determine the product over all phases, but not the selected phase when* $|\Omega| > 1$.

Proof. Diagonal translation has free orbits of size R on the tuples (j_i). Frobenius cycles the factors in each orbit, so each orbit product lies in $K[Z]$, and partitioning the tuples proves the factorization. The zero-phase factors are exactly the M/K-conjugates of u, giving the norm. On the p-Kummer quotient the norm exponent is

$$1 + Q + \cdots + Q^{R-1} \equiv R \pmod{p}.$$

Since $p \nmid R$, multiplication by R is invertible. Independent Frobenius shifts of the components preserve every separate minimal polynomial while translating the phase, which proves the final limitation. □

In the multicomponent part of Z, the relative degrees supplied by Lenstra's component theorem satisfy the coprimality hypothesis in Proposition 4.2. Thus Z is an order assertion about one distinguished factor, not about the unmarked additive composed product.

4.2 The singleton-even Conway–Fermat problem

Put

$$h_n = 2^{n+1}, \qquad c_n = \kappa_{h_n} = 2^{2^n}, \qquad E_n = \mathbb{F}_{2^{h_n}}, \qquad F_n = 2^{2^n} + 1,$$

and take $E_{-1} = \mathbb{F}_2$. The Conway tower relation is

$$c_n^2 + c_n = a_{n-1}, \qquad a_{n-1} = \prod_{j=0}^{n-1} c_j.$$

Let

$$\delta_n = \mathrm{ord}\big(c_n E_{n-1}^\times\big) \quad \text{in } E_n^\times / E_{n-1}^\times.$$

Proposition 4.3 (Conway–Fermat form)**.** *One has* $1 < \delta_n \mid F_n$*. The zero arm at every prime divisor of* F_n *is equivalent to*

$$\delta_n = F_n.$$

Equivalently, $c_0 c_n$ *is primitive in* E_n *for every* n *if and only if these equalities hold at every preceding level. This is the selected-order problem studied for Conway towers by Popovych [11].*

Proof. The quotient group has order

$$\frac{2^{2^{n+1}} - 1}{2^{2^n} - 1} = F_n,$$

and $c_n \notin E_{n-1}$, so its quotient order is a nontrivial divisor. Every odd prime $p \mid F_n$ has $f(p) = 2^{n+1}$, and its entire primary part in $E_n^\times$ lies in the quotient. Proposition 2.1 therefore gives $m_p = 0$ exactly when the quotient order contains the full p-part. Imposing this at every prime divisor of F_n is equivalent to $\delta_n = F_n$. The primitive-product equivalence follows because the successive quotient orders are supported on pairwise coprime Fermat numbers and c_0 has order three. □

The precise zero-at-multiples coordinate is especially small. Define Fibonacci polynomials over $\mathbb{F}_2$ by

$$S_0(X) = 0, \qquad S_1(X) = 1, \qquad S_{j+2}(X) = S_{j+1}(X) + XS_j(X).$$

Proposition 4.4 (zero-at-multiples normal form)**.** *For every* $n \geq 0$*,*

$$\delta_n = \min\{j > 0 : S_j(a_{n-1}) = 0\}.$$

In fact,

$$S_j(a_{n-1}) = 0 \quad \Longleftrightarrow \quad \delta_n \mid j.$$

Consequently,

$$\delta_n = F_n \quad \Longleftrightarrow \quad S_{F_n/\ell}(a_{n-1}) \neq 0 \quad \textit{for every prime } \ell \mid F_n.$$

Proof. Put $c = c_n$, $a = a_{n-1} = c(c+1)$, and $u = (c+1)/c$. Induction from the recurrence gives the Binet identity

$$S_j(a) = c^j + (c+1)^j = c^j(1 + u^j).$$

The quotient-to-norm-one isomorphism

$$E_n^\times / E_{n-1}^\times \longrightarrow \ker \mathrm{N}_{E_n/E_{n-1}}, \qquad zE_{n-1}^\times \longmapsto z^{|E_{n-1}^\times|},$$

sends the class of c to u. Thus $\mathrm{ord}(u) = \delta_n$, and the displayed equivalences follow. □

The recurrence is a lossless encoding of the unknown order, not an order bound. Its least zero period is already δ_n. In particular, factorizations, trace identities, or automata derived only from this word reconstruct the selected order rather than force it to be F_n. The classical iterated quadratic tower of Wiedemann has a related norm-one form but a different recursion, so its order results do not transfer by a Frobenius identification [15]. The recent general Artin–Schreier tower theorem of Cagliero, Herman, and Szechtman proves the exact product of the relative orders, but its maximality condition remains precisely that every relative factor δ_j be full; it does not prove $\delta_j = F_j$ [2].

There is nevertheless an exact elliptic realization of this arm. It is useful because it isolates, rather than removes, the remaining Kummer coordinate.

Proposition 4.5 (supersingular realization and pairing boundary)**.** *On*

$$\mathcal{E}: \quad y^2 + y = x^3 + x^2$$

put $P_n = (c_n, c_{n+1})$*. Then* $P_n \in \mathcal{E}$*, and for* $n \geq 2$

$$[F_n]P_n = O.$$

More precisely, if π *is absolute Frobenius and* $\iota = \pi + [1]$*, then*

$$\iota(x, y) = (x + 1, y + x + 1), \qquad \iota^2 = [-1], \qquad \pi^4 = [-4].$$

For

$$g = \frac{y+1}{y}$$

one has

$$g(P_n) = \frac{c_{n+1} + 1}{c_{n+1}}, \qquad \operatorname{ord}(g(P_n)) = \delta_{n+1}.$$

If $T = (0, 0)$*, then* T *has order five and*

$$\operatorname{div}(g) = 2[-T] + [2T] - 2[T] - [-2T].$$

Thus g *is the quotient of the two fixed doubling Miller functions at* $-T$ *and* T*, but it is not a Weil or reduced-Tate pairing function attached to the* F_n*-torsion point* P_n*.*

Proof. The Conway recursion gives

$$c_{n+1}^2 + c_{n+1} = a_n = c_n(c_n^2 + c_n) = c_n^3 + c_n^2,$$

so P_n lies on $\mathcal{E}$. The displayed coordinate formula for $\iota(P) = \pi(P) + P$ follows from the chord law; applying it twice gives the inverse $(x, y) \mapsto (x, y + 1)$. Hence $\pi^2 + 2\pi + 2 = 0$ in $\operatorname{End}(\mathcal{E})$, and squaring this relation gives $\pi^4 = [-4]$.

Put $H = 2^{n+1}$. The 2^H-Frobenius fixes c_n and sends c_{n+1} to $c_{n+1} + 1$, so it sends P_n to $-P_n$. For $n \geq 2$,

$$\pi^H = (\pi^4)^{2^{n-1}} = [2^{2^n}],$$

and therefore $[2^{2^n} + 1]P_n = O$.

The formula for $g(P_n)$ and Proposition 4.4 give its order. The five rational points of $\mathcal{E}/\mathbb{F}_2$ show that T has order five. Direct intersection with the two horizontal lines gives

$$\operatorname{div}(y) = 2[T] + [-2T] - 3[O], \qquad \operatorname{div}(y + 1) = 2[-T] + [2T] - 3[O],$$

and subtraction gives the divisor of g. Equivalently, $g = f_{2,-T}/f_{2,T}$ for the $\mathbb{F}_2$-rational functions normalized at O. Its divisor has coefficients $2, 1, -2, -1$ at fixed rational five-torsion, so it cannot be an F_n-Miller function attached to P_n, even modulo an F_n-th power. Any Weil or reduced Tate pairing on $\mathcal{E}[F_n]$ takes values in μ_{F_n}, whereas $g(P_n)$ has order dividing F_{n+1}. The Fermat numbers are pairwise coprime and $\delta_{n+1} > 1$, so these values cannot agree. $\square$

The proposition supplies exact CM annihilator control of the literal Conway point, but generator status still does not follow from annihilation by F_n. For $n \geq 3$, no elliptic endomorphism can carry P_n to P_{n-1}: the order of such an image would divide both coprime numbers F_n and F_{n-1}, although $P_{n-1} \neq O$. For $n = 2$, the same conclusion follows from $\operatorname{ord}(P_1) = 3$ and $\operatorname{ord}(P_2) \mid 17$. The actual ancestry relation $x(P_n) = y(P_{n-1})$ is therefore not an isogeny division tower. The unresolved statement remains exactly the next-level value $\operatorname{ord}(g(P_n)) = F_{n+1}$.

Proposition 4.6 (two complete multicomponent levels)**.** *Assertion Z holds at $h = 12$ and $h = 24$.*

Proof. At $h = 12$, the selected element is the sum of components of degrees six and four; take

$$(A, B, J) = (\mathbb{F}_{2^6}, \mathbb{F}_{2^4}, \mathbb{F}_{2^2}).$$

At $h = 24$, the components have degrees six and eight, but the relevant maximal proper support fields are

$$(A, B, J) = (\mathbb{F}_{2^{12}}, \mathbb{F}_{2^8}, \mathbb{F}_{2^4}).$$

The degree-six component lies in $A \setminus J$, while the degree-eight component lies in $B \setminus J$. In both cases $[A : J] = 3$ and $[B : J] = 2$, so multiplication identifies $A \otimes_J B$ with their compositum. The selected sum corresponds to $x \otimes 1 + 1 \otimes y$, a tensor of rank two, whereas every element of $A^\times B^\times$ has tensor rank one. Hence the selected class is nontrivial in

$$\mathbb{F}_{2^h}^\times/(A^\times B^\times).$$

The quotient orders are, respectively,

$$\frac{(2^{12} - 1)(2^2 - 1)}{(2^6 - 1)(2^4 - 1)} = \Phi_{12}(2) = 13, \qquad \frac{(2^{24} - 1)(2^4 - 1)}{(2^{12} - 1)(2^8 - 1)} = \Phi_{24}(2) = 241.$$

Both are prime, so a nontrivial class generates the complete primitive-support quotient. Proposition 4.1 gives Z at the two levels. □

The universal arm Z therefore remains the simultaneous generation problem for the distinguished synchronized norm, including the all-level Conway–Fermat equalities $\delta_n = F_n$.

5 The ordinary odd Kummer arm

Fix an odd prime r, let $b = b_r = d(\alpha_r)$, and set

$$x_0 = \alpha_r, \qquad x_a = \kappa_{r^a} \ (a \geq 1), \qquad y_a = x_a + 1, \qquad F_a = \mathbb{F}_{2^{r^a b}}.$$

The Kummer equations are $x_1^r = \alpha_r$ and $x_a^r = x_{a-1}$ for $a > 1$.

Proposition 5.1 (projective order criterion)**.** *For every $a \geq 1$,*

$$\mathrm{N}_{F_a/F_{a-1}}(y_a) = y_{a-1}.$$

The cyclic quotient $G_{r,a} = F_a^\times/F_{a-1}^\times$ has order

$$L_{r,a} = \frac{2^{r^a b} - 1}{2^{r^{a-1} b} - 1} = \Phi_{r^a}(2^b).$$

If $p \neq r$ is a prime divisor of $L_{r,a}$, then

$$\mathrm{ord}_p(2^b) = r^a, \qquad f(p) = r^a s \text{ for some } s \mid b, \qquad \mathcal{Q}(f(p)) = \{r^a\}.$$

Conversely, every singleton row with component r^a arises from such a prime. For each corresponding row,

$$m_p = 1 \quad\Longleftrightarrow\quad v_p\big(\mathrm{ord}(y_a F_{a-1}^\times)\big) = v_p(L_{r,a}).$$

Proof. The relative conjugates of x_a are ξx_a, $\xi \in \mu_r$. Evaluating their binomial at one gives the norm identity. The quotient order follows from the field degrees. For $p \neq r$, the cyclotomic divisor criterion applied to $p \mid \Phi_{r^a}(2^b)$ gives $\mathrm{ord}_p(2^b) = r^a$. Writing $h = f(p)$ then gives $h = r^a s$ with $s \mid b$, and Lenstra's minimality and component theorem give $\mathcal{Q}(h) = \{r^a\}$. The converse is the same argument in reverse.

The prime p does not divide $|F_{a-1}^\times|$, so its full primary contribution lies in the quotient. The selected element x_a has order prime to p, hence is a p-th power; therefore $m_p \geq 1$. Proposition 2.1 applied to y_a proves the final equivalence. ☐

For a particular row $h = r^a s$, $s \mid b$, the large ambient test may be reduced to the transverse field $K = \mathbb{F}_{2^{r^a s}}$:

$$\Theta_{r,a,s} = \mathrm{N}_{F_a/K}(x_a + 1).$$

Then $m_p = 1$ exactly when the order of $\Theta_{r,a,s}$ contains the full p-part of $2^{r^a s} - 1$. By contrast,

$$\mathrm{N}_{F_a/F_{r,0}}(x_a + 1) = 1 + \alpha_r$$

lies in a group whose order is prime to p. The lower Kummer norm is therefore blind to the current primary coordinate; the transverse norm is not.

5.1 The marked cyclotomic unit

Let $N_a = \mathrm{ord}(x_a)$, choose a primitive complex N_a-th root ξ_{N_a}, and choose a prime $\mathfrak{P}$ above two whose residue map sends ξ_{N_a} to the selected element x_a. Put $\varepsilon_a = 1 - \xi_{N_a}$. Its reduction is $1 + x_a = y_a$.

Proposition 5.2 (selected residue-symbol form)**.** *Let $p \neq r$ be a prime divisor of $L_{r,a}$. In a cyclotomic field containing μ_{N_a} and μ_p, the following are equivalent:*

1. *the projective class of y_a has full p-primary order;*
2. *y_a is not a p-th power in F_a;*
3. *the power-residue symbol satisfies*
$$\left(\frac{\varepsilon_a}{\mathfrak{P}}\right)_p \neq 1;$$
4. *$\mathfrak{P}$ does not split in the Kummer extension obtained by adjoining a p-th root of ε_a.*

Proof. The first two conditions are Proposition 5.1. The usual unramified power-residue formula is

$$\left(\frac{\varepsilon_a}{\mathfrak{P}}\right)_p = \overline{\varepsilon_a}^{(|F_a|-1)/p}.$$

It is nontrivial exactly when the residue is not a p-th power. Kummer theory identifies this symbol with the Frobenius action on a chosen p-th root; see [14]. ☐

This form explains the real difficulty of O. The conductor and the unmarked conjugate set are determined by the tower, but the conjecture asks for one residue symbol at the prime over two selected by Conway ancestry.

Proposition 5.3 (the selected prime is a unique inert ray)**.** *Put $M = \mathrm{ord}(\alpha_r)$. Choose compatible primitive roots ξ_{N_a} with $\xi_{N_0} \mapsto \alpha_r$ and $\xi_{N_a}^r = \xi_{N_{a-1}}$. Then*

$$N_a = \mathrm{ord}(x_a) = r^a M, \qquad \mathrm{ord}_{N_a}(2) = br^a.$$

Let $K_a = \mathbb{Q}(\xi_{N_a})$, and let $\mathfrak{P}_a \mid 2$ be the prime selected by $\xi_{N_a} \mapsto x_a$. For $a \geq 1$, $\mathfrak{P}_a$ is the unique prime of K_a above $\mathfrak{P}_{a-1}$, with relative residue degree r. Moreover,

$$\mathrm{N}_{K_a/K_{a-1}}(1 - \xi_{N_a}) = 1 - \xi_{N_{a-1}}.$$

If $p \neq r$ divides $L_{r,a}$, then the lower residue field has trivial p-Kummer quotient. Consequently the residue-field norm

$$F_a^\times/(F_a^\times)^p \longrightarrow F_{a-1}^\times/(F_{a-1}^\times)^p,$$

or equivalently the corresponding finite-field corestriction, sends the selected current class to zero and imposes no restriction on it.

Proof. The irreducible Kummer steps $X^r - x_{a-1}$ increase the r-primary part of the root order by one and preserve every other primary part. This gives $N_a = r^a M$. The defining-field equality $\mathbb{F}_2(x_a) = \mathbb{F}_{2^{br^a}}$ then gives $\mathrm{ord}_{N_a}(2) = br^a$.

The first irreducible binomial also forces $r \mid M$. Hence $[K_a : K_{a-1}] = r$, while the displayed orders show that the residue-degree ratio at the selected primes is also r. Since two is unramified in these odd-conductor cyclotomic fields, the fundamental identity forces one prime above $\mathfrak{P}_{a-1}$, of residue degree r. Taking the product over the r-th roots of unity gives the norm identity.

Finally, $p \mid \Phi_{r^a}(2^b)$ but $p \nmid 2^{br^{a-1}} - 1$. Raising to the p-th power is therefore an automorphism of the lower residue-field multiplicative group, so its p-Kummer quotient is zero. The last assertion follows after reduction at $\mathfrak{P}_{a-1}$; no claim is made that the global cyclotomic-unit class has trivial number-field norm. □

Thus full ancestry leaves no hidden choice of a prime after the birth level: it determines one compatible inert ray. What it does not determine is the new augmentation coordinate of the cyclotomic unit on that ray. Global reciprocity only relates the product of the conjugate symbols, which is one, and ramified conductor formulas at primes above p determine a different local character. A nontrivial ray character may still vanish on the fixed class of $\mathfrak{P}_a$. The missing theorem is therefore a selected-prime Artin-symbol nonvanishing statement, not a decomposition, norm, or conductor calculation.

Proposition 5.4 (phase conservation above the birth level)**.** *Fix a, a prime $p \neq r$ dividing $L_{r,a}$, and for $j \geq a$ put*

$$\Theta_j = (1 + x_j)^{(2^{r^j b} - 1)/p} \in \mu_p.$$

Then

$$\Theta_j = \Theta_a \qquad (j \geq a).$$

Thus passage to later levels of the same Kummer spine creates no new current p-coordinate.

Proof. Let

$$S_j = \frac{2^{r^j b} - 1}{2^{r^{j-1} b} - 1}.$$

The norm identity gives $(1 + x_j)^{S_j} = 1 + x_{j-1}$, while

$$\frac{2^{r^j b} - 1}{p} = S_j \frac{2^{r^{j-1} b} - 1}{p}.$$

Therefore $\Theta_j = \Theta_{j-1}$, and induction proves the claim. □

For $r \neq 3$, assertion O is exactly the universal nontriviality of these marked symbols at their birth levels. General lower bounds for orders in finite fields and cyclotomic constructions do not by themselves evaluate the selected residue symbol.

6 The cubic-power arm

Let $h = 3^k$, $\zeta_k = \kappa_{3^k}$, and $\gamma_k = \zeta_k + \zeta_k^{-1}$.

Proposition 6.1 (cyclotomic and half-angle structure)**.** *The element ζ_k is a primitive 3^{k+1}-st root of unity,*

$$\mathbb{F}_2(\zeta_k) = \mathbb{F}_{2^{2h}}, \qquad \mathbb{F}_2(\gamma_k) = \mathbb{F}_{2^h}.$$

If s is the inverse of two modulo 3^{k+1}, then

$$\zeta_k + 1 = \zeta_k^s(\zeta_k^s + \zeta_k^{-s}), \qquad (\zeta_k^s + \zeta_k^{-s})^2 = \gamma_k,$$

and hence

$$\operatorname{ord}(\zeta_k + 1) = 3^{k+1} \operatorname{ord}(\gamma_k).$$

The translates $\zeta_k + 2$ and $\zeta_k + 3$ have the same order. Consequently, for a prime p with $f(p) = 3^k$, one has

$$m_p = 1 \iff v_p(\operatorname{ord}(\gamma_k)) = v_p(2^h - 1).$$

Assuming γ_{k-1} is primitive, all such rows have excess one exactly when γ_k is primitive.

Proof. The defining Kummer equation gives $\zeta_k^{3^k} = \kappa_2$, a primitive cube root. Since two has order $2 \cdot 3^k$ modulo 3^{k+1}, the cyclotomic polynomial $X^{2h} + X^h + 1$ is irreducible over $\mathbb{F}_2$. The least exponent $j > 0$ with $2^j \equiv \pm 1 \pmod{3^{k+1}}$ is h, which gives the degree of γ_k.

The half-angle identities are direct. Their two factors lie in the coprime circle and real parts of $\mathbb{F}_{2^{2h}}^\times$, of orders dividing $2^h + 1$ and $2^h - 1$, respectively. Squaring preserves odd order, so the order formula follows. The other two translates are Frobenius conjugate to the same decomposition. Finally, a prime with $f(p) = h$ cannot divide the 3-power factor, so Proposition 2.1 gives the stated criterion. □

The selected Gaussian period obeys

$$\gamma_k^3 + \gamma_k = \gamma_{k-1}, \qquad \mathrm{N}_{\mathbb{F}_{2^{3^k}}/\mathbb{F}_{2^{3^{k-1}}}}(\gamma_k) = \gamma_{k-1}.$$

This propagates every old primary factor. It does not control the new factor $\Phi_{3^k}(2)$.

Proposition 6.2 (selected norm-one generator)**.** *Put*

$$q_k = 2^{3^{k-1}}, \qquad U_k = \ker\left(\mathrm{N}_{\mathbb{F}_{q_k^3}/\mathbb{F}_{q_k}}\right), \qquad \eta_k = \gamma_k^{q_k - 1},$$

and set $\eta_0 = 0$. Then

$$|U_k| = q_k^2 + q_k + 1 = \Phi_{3^k}(2), \qquad \mathbb{F}_2(\eta_k) = \mathbb{F}_{2^{3^k}},$$

and η_k is the selected root of the irreducible cubic

$$X^3 + \eta_{k-1}X^2 + (\eta_{k-1} + 1)X + 1.$$

Equivalently,

$$\mathrm{N}(\eta_k) = 1, \qquad \mathrm{Tr}(\eta_k) = \eta_{k-1}, \qquad e_2(\eta_k) = \eta_{k-1} + 1.$$

Moreover,

$$C_k \quad \Longleftrightarrow \quad C_{k-1} \text{ and } \mathrm{ord}(\eta_k) = q_k^2 + q_k + 1,$$

where C_j denotes primitivity of γ_j.

Proof. The element η_k has norm one by its definition. A direct Frobenius calculation from $\zeta_k^{q_k}$ gives its displayed cubic; its symmetric coefficients give the norm, trace, and second elementary symmetric function. It cannot lie in the base field, since a base-field element of U_k has order dividing $\gcd(q_k - 1, 3) = 1$. Hence the cubic is irreducible. Since its quadratic coefficient recovers η_{k-1}, induction from $\eta_0 = 0$ gives $\mathbb{F}_2(\eta_k) = \mathbb{F}_{2^{3^k}}$.

The cyclic group $\mathbb{F}_{q_k^3}^\times$ splits as the product of $\mathbb{F}_{q_k}^\times$ and U_k, whose orders are coprime because $q_k \equiv 2 \pmod 3$. The norm determines the old factor of γ_k, while exponentiation by $q_k - 1$ detects the new U_k-factor without changing its order. This proves the last equivalence. $\square$

Proposition 6.3 (exact Capelli dichotomy)**.** *Let $Q_k \in \mathbb{F}_2[X]$ be the minimal polynomial of η_k, and let ℓ be a prime divisor of $N_k = \Phi_{3^k}(2)$. Exactly one of the following occurs:*

1. *$\eta_k \notin (\mathbb{F}_{2^{3^k}}^\times)^\ell$, and $Q_k(X^\ell)$ is irreducible of degree $3^k\ell$;*
2. *$\eta_k \in (\mathbb{F}_{2^{3^k}}^\times)^\ell$, and $Q_k(X^\ell)$ is the product of exactly ℓ distinct irreducible polynomials, each of degree 3^k.*

In particular, both the successful and failed cases have an odd number of irreducible factors. Factor-count parity, and hence any criterion which sees only that parity, cannot decide the current C-coordinate.

Proof. Put $h = 3^k$. The preceding proposition gives $\mathbb{F}_2(\eta_k) = \mathbb{F}_{2^h}$, while $\ell \mid 2^h - 1$ gives $\mu_\ell \subset \mathbb{F}_{2^h}$. Capelli's lemma and the prime-degree Kummer criterion therefore say that $Q_k(X^\ell)$ is irreducible exactly when η_k is not an ℓ-th power.

Suppose instead that $b^\ell = \eta_k$ in $\mathbb{F}_{2^h}$. Every root of $Q_k(X^\ell)$ is an ℓ-th root of a Frobenius conjugate of η_k, hence belongs to $\mathbb{F}_{2^h}$, because that field also contains μ_ℓ. Such a root has degree at least h over $\mathbb{F}_2$, since its ℓ-th power has degree h, and therefore has degree exactly h. The composition is separable and has degree $h\ell$, so it has exactly ℓ distinct irreducible factors, all of degree h. Finally ℓ is odd because it divides $2^h - 1$. $\square$

For a prime $\ell \mid |U_k|$, put

$$\Omega_{k,\ell} = \eta_k^{|U_k|/\ell}.$$

The exact terminal test is

$$\mathrm{ord}(\eta_k) = |U_k| \quad \Longleftrightarrow \quad \Omega_{k,\ell} \neq 1 \quad \text{for every prime } \ell \mid |U_k|.$$

This formulation again retains the full ℓ-primary valuation. The minimal polynomial of γ_k is also explicit. If $D_n(X, 1)$ is the Dickson polynomial defined by $D_n(t + t^{-1}, 1) = t^n + t^{-n}$, then

$$P_k(X) = D_{3^k}(X, 1) + 1 = (X^3 + X)^{\circ k} + 1$$

is irreducible of degree 3^k, and C_k says precisely that it is a primitive polynomial. Irreducibility is therefore already built into the tower; primitivity is the open statement. General lower bounds for multiplicative orders of Gaussian periods, such as [1], do not give maximal order for this recursively selected period.

In fact degree is maximally insensitive here. Every prime $\ell \mid \Phi_{3^k}(2)$ satisfies $\mathrm{ord}_\ell(2) = 3^k$. If $z \neq 1$ lies in U_k, choose such a prime $\ell \mid \mathrm{ord}(z)$. Then $[\mathbb{F}_2(z) : \mathbb{F}_2]$ is both divisible by $\mathrm{ord}_\ell(2) = 3^k$ and at most 3^k, hence equals 3^k. Thus even an element of proper order in U_k has the same full field degree as a generator.

The recursion in Proposition 6.2 is lossless. It reconstructs every finite selected truncation up to Frobenius, and Frobenius preserves multiplicative order. Trace, norm, and the sparse cubic therefore identify the correct selected coordinate but do not evaluate $\Omega_{k,\ell}$. This is the exact remaining content of arm C.

7 The exceptional arm

Let $h = 3^k$, $q = 2^h$, $\zeta = \zeta_k$, and $\omega = \zeta^h \in \mathbb{F}_4$.

Proposition 7.1 (unconditional lower bound)**.** *If p is a prime with $f(p) = 2 \cdot 3^k$, then $m_p \geq 4$.*

Proof. Such a prime divides $2^h + 1$ and divides neither 3^{k+1} nor $2^h - 1$. The element ζ has order 3^{k+1}, while Proposition 6.1 gives

$$\mathrm{ord}(\zeta + m) = 3^{k+1}\,\mathrm{ord}(\gamma_k) \quad (m = 1, 2, 3),$$

with $\mathrm{ord}(\gamma_k) \mid 2^h - 1$. Hence all four translates $m = 0, 1, 2, 3$ have order prime to p, and each is a p-th power in its defining field. The definition of the excess gives the claim. $\square$

The translate by four lies in the compositum with $\mathbb{F}_{16}$, so its first useful invariant is a quadratic norm rather than an order calculation in $\mathbb{F}_{2^{2h}}$.

Proposition 7.2 (corrected norm)**.** *In $\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}}$, the nontrivial relative Frobenius sends the finite nimber 4 to 5, and*

$$\mathrm{N}_{\mathbb{F}_{2^{4h}}/\mathbb{F}_{2^{2h}}}(\zeta + 4) = (\zeta + 4)(\zeta + 5) = \zeta^2 + \zeta + \omega = A_k.$$

Proof. The intersection of $\mathbb{F}_{16}$ and $\mathbb{F}_{2^{2h}}$ is $\mathbb{F}_4$, so the relative Frobenius is the nontrivial automorphism of $\mathbb{F}_{16}/\mathbb{F}_4$ and exchanges 4 and 5. These are the two roots of $Y^2 + Y + \omega$; hence $4 \cdot 5 = 4^2 + 4 = \omega$. Expanding the product proves the formula. $\square$

Define

$$R_k(Y) = \mathrm{Res}_X\left(X^{2h} + X^h + 1,\ Y + X^h + X^2 + X\right) \in \mathbb{F}_2[Y].$$

Theorem 7.3 (exact exceptional target)**.** *The polynomial R_k is the irreducible degree-$2h$ minimal polynomial of A_k. The prime divisors of*

$$\Psi_k = \frac{\Phi_{2\cdot 3^k}(2)}{3}$$

are exactly the primes $\ell \neq 3$ with $\mathrm{ord}_\ell(2) = 2h$, with their full valuations in $2^{2h} - 1$. The following are equivalent:

1. *$m_\ell = 4$ for every prime ℓ with $f(\ell) = 2 \cdot 3^k$;*
2. *$\Psi_k \mid \mathrm{ord}(M_k)$, where $M_k = A_k^{q-1}$;*

3. *A_k is not an ℓ-th power in $\mathbb{F}_{2^{2h}}$ for every prime $\ell \mid \Psi_k$;*

4. *$R_k(X^\ell)$ is irreducible over $\mathbb{F}_2$ for every prime $\ell \mid \Psi_k$.*

Proof. The element ζ has minimal polynomial $X^{2h}+X^h+1$. Suppose $A_k^{2^j} = A_k$, and reduce $a = 2^j$ modulo $3h$. The resulting relation on ζ has support the symmetric difference of

$$\{ah, a, 2a\} \quad \text{and} \quad \{h, 1, 2\}.$$

Every multiple of $X^{2h} + X^h + 1$ of degree below $3h$ has support a disjoint union of triples $\{r, r+h, r+2h\}$. If $a \equiv 2 \pmod 3$, the support contains $h, 2h$ but not zero. If $a \equiv 1 \pmod 3$, cancellation of the h-terms leaves at most two members in each residue class modulo h. Both are impossible for a nonempty union of triples. The support is therefore empty, forcing $a = 1$. Thus A_k has Frobenius orbit length $2h$, and the displayed resultant is its irreducible minimal polynomial.

This also verifies the defining field used in Proposition 7.2. Put $\beta = \zeta + 4$. The identity $A_k = \mathrm{N}(\beta)$ puts A_k in $\mathbb{F}_2(\beta)$, so the degree just proved shows that $\mathbb{F}_{2^{2h}} \subseteq \mathbb{F}_2(\beta)$. On the other hand $\beta \notin \mathbb{F}_{2^{2h}}$: otherwise $4 = \beta + \zeta$ would belong to that field, contrary to $\mathbb{F}_{16} \cap \mathbb{F}_{2^{2h}} = \mathbb{F}_4$. Hence $\mathbb{F}_2(\beta) = \mathbb{F}_{2^{4h}}$.

Writing $a_0 = 2^{3^{k-1}}$, one has

$$\Phi_{2\cdot 3^k}(2) = a_0^2 - a_0 + 1 \equiv 3 \pmod 9.$$

After removing its single factor of three, the cyclotomic divisor criterion gives the asserted prime support and valuations.

Fix $\ell \mid \Psi_k$. Proposition 7.1 gives $m_\ell \geq 4$. Propositions 2.3 and 7.2 give

$$m_\ell = 4 \quad \Longleftrightarrow \quad A_k \notin (\mathbb{F}_{2^{2h}}^\times)^\ell.$$

Raising to $q-1$ is invertible on the current ℓ-Kummer quotient, because $q-1 \equiv -2 \pmod \ell$. Hence this is equivalent to the full ℓ-primary part appearing in $\mathrm{ord}(M_k)$. Imposing it for every current prime gives assertion 2. Finally, μ_ℓ lies in $\mathbb{F}_{2^{2h}}$, so Kummer theory and Capelli's lemma identify non-ℓ-power of A_k with irreducibility of $R_k(X^\ell)$. $\square$

Proposition 7.4 (effective bound on the exceptional column)**.** *Let $k \geq 2$, let $h = 3^k$, and let $\ell \mid \Psi_k$ be prime. Then*

$$m_\ell < 16h^4.$$

More precisely, let t be the least power of two satisfying

$$t \geq \lceil \log_2(h-1) \rceil .$$

Then $m_\ell \leq 4^t - 1$.

Proof. Put

$$K = \mathbb{F}_{4^h} = \mathbb{F}_4(\zeta), \qquad C_a = \mathbb{F}_{4^t}, \qquad L_a = KC_a = \mathbb{F}_{4^{ht}},$$

where $t = 2^{a-1}$. Choose a character $\chi : K^\times \to \mu_\ell$ of exact order ℓ, and define

$$\chi_a = \chi \circ \mathrm{N}_{L_a/K}, \qquad S_{a,k,\ell} = \sum_{c \in C_a} \chi_a(\zeta + c).$$

Because h is odd and t is a power of two, the fields K and C_a are linearly disjoint over $\mathbb{F}_4$, and $[L_a : K] = t$. This degree is prime to ℓ, so norm induces an isomorphism on ℓ-Kummer quotients.

In particular, χ_a has kernel $(L_a^\times)^\ell$. No term $\zeta + c$ vanishes, and equality in the complex triangle inequality gives

$$\zeta + C_a \subseteq (L_a^\times)^\ell \quad \iff \quad S_{a,k,\ell} = 4^t.$$

The character χ is the primitive even Dirichlet character modulo $X^h + \omega$ over $\mathbb{F}_4[X]$. Write its L-polynomial as

$$L_\chi(T) = (1-T)\prod_{j=1}^{h-2}(1-\alpha_j T).$$

The function-field Riemann hypothesis gives $|\alpha_j| = 2$, and constant-field extension raises each reciprocal root to its t-th power [12, Chapters 4, 8, and 9]. Comparing the degree-one coefficient after extension gives

$$S_{a,k,\ell} = -\left(1 + \sum_{j=1}^{h-2}\alpha_j^t\right), \qquad |S_{a,k,\ell}| \le 1 + (h-2)2^t.$$

If $2^t \ge h-1$, then

$$1 + (h-2)2^t < 4^t.$$

Thus some $c \in C_a$ makes $\zeta + c$ a non-ℓ-th power. Kummer-quotient injectivity from its defining field to L_a shows that the same is true in the defining field. The elements of C_a are exactly the finite nimbers below 4^t, so $m_\ell \le 4^t - 1$.

Finally, the least power of two t above $\lceil \log_2(h-1)\rceil$ satisfies

$$t < 2(\log_2 h + 1),$$

and hence $4^t < 16h^4$. This proves the stated bound. □

The open statement D is therefore the family

$$D_k': \qquad \Psi_k \mid \operatorname{ord}(M_k).$$

It is a current-level assertion, not a cumulative divisibility statement. The factor A_k in Proposition 7.2 is also essential: replacing it by a scalar product of separate translates gives the wrong norm.

Proposition 7.5 (prime current factor)**.** *Let $k \ge 2$, put*

$$s = 3^{k-1}, \qquad a = 2^s, \qquad \Lambda_k = a^2 - a + 1 = 3\Psi_k,$$

and define $P_k = M_k^{a+1}$. Then

$$P_k^{\Lambda_k} = 1, \qquad [\mathbb{F}_2(P_k) : \mathbb{F}_2] = 2 \cdot 3^k,$$

and exponentiation by $a+1$ preserves every primary part supported on Ψ_k. Consequently D_k' holds whenever Ψ_k is prime. In particular, $\Psi_2 = 19$ and $\Psi_3 = 87211$ are prime, so D_2' and D_3' hold.

Proof. Write $h = 3^k$ and $q = a^3 = 2^h$. Since

$$q + 1 = (a+1)(a^2 - a + 1),$$

the norm-one relation $M_k^{q+1} = 1$ gives $P_k^{\Lambda_k} = 1$. Moreover, $\gcd(\Lambda_k, a+1) = 3$ and $v_3(\Lambda_k) = 1$, hence $\gcd(\Psi_k, a+1) = 1$. This proves the assertion about primary parts.

The congruences

$$a \equiv h - 1, \qquad a^2 \equiv h + 1, \qquad a^3 \equiv -1, \qquad a^4 \equiv 2h + 1 \pmod{3h}$$

give, by direct Frobenius expansion,

$$P_k = \frac{\zeta^4 + \zeta + \omega}{\omega^2 \zeta^4 + \zeta^3 + 1}.$$

Also $P_k^q = P_k^{-1}$. If $P_k \in \mathbb{F}_q$, then $P_k = 1$; cross-multiplication would give

$$\zeta^{h+4} + \zeta^h + \zeta^3 + \zeta + 1 = 0.$$

If $P_k \in \mathbb{F}_{a^2}$, its order divides $\gcd(\Lambda_k, a^2 - 1) = 3$, so $P_k \in \{1, \omega, \omega^2\}$. The two remaining values similarly give

$$\zeta^{h+3} + \zeta = 0$$

or

$$\zeta^{h+4} + \zeta^{h+3} + \zeta^4 + \zeta^3 + \zeta + 1 = 0.$$

Each displayed polynomial is nonzero and has degree below $2h$, contradicting the minimal polynomial $X^{2h} + X^h + 1$ of ζ.

Every proper subfield of $\mathbb{F}_{2^{2h}}$ lies in $\mathbb{F}_q$ or $\mathbb{F}_{a^2}$. Thus P_k has degree $2h$. If Ψ_k is prime, its order, a divisor of $3\Psi_k$, cannot divide three and must therefore contain Ψ_k. The primary-part assertion transfers this divisibility to $\operatorname{ord}(M_k)$. $\square$

The cubic and exceptional current factors are coprime:

$$\gcd\big(\Phi_{3^k}(2), \Phi_{2\cdot 3^k}(2)\big) = 1.$$

Thus arm C and arm D occupy distinct primary coordinates. Norm recursion in the cubic tower cannot by itself transfer the missing order from one arm to the other. Their common Conway ancestry is relevant only through the explicitly selected additive coefficients.

Proposition 7.6 (common toric cubic dynamics)**.** *Over a field of characteristic two, for $z \neq 0$ put*

$$\varpi(z) = z + z^{-1}, \qquad D(X) = X^3 + X, \qquad F_c(U) = U^3 + cU^2 + cU,$$

where $c^2 + c = 1$. Then

$$D(\varpi(z)) = \varpi(z^3), \qquad F_c(U) = D(U + c) + c^2,$$

and

$$F_{c^2}(F_c(U)) = D(D(U + c)) + c.$$

In the literal cubic tower choose alternating $c_j \in \mathbb{F}_4$ with $c_j^2 + c_j = 1$ and $c_{j-1} = c_j^2$, and put $u_j = \gamma_j + c_j$. Then

$$F_{c_j}(u_j) = u_{j-1}.$$

Consequently the selected C ancestry is the inversion quotient of the 3-division tree on $\mathbb{G}_m$, while this auxiliary recursion used in the analysis of D is a two-periodic affine coordinate on the same toric tree. This does not identify u_k with the exceptional target A_k, whose Kummer class remains the one in Theorem 7.3.

Proof. Expanding $(z+z^{-1})^3+(z+z^{-1})$ in characteristic two leaves z^3+z^{-3}. The second identity is another direct expansion using $c^2+c=1$, which also gives $c^3=1$. Applying it first with c and then with c^2, and using $c^4=c$, gives the third identity. Finally, $\gamma_j^3+\gamma_j=\gamma_{j-1}$, so the second identity gives

$$F_{c_j}(u_j)=D(\gamma_j)+c_j^2=\gamma_{j-1}+c_{j-1}=u_{j-1}.$$

□

This exact identification rules out direct identification with several tempting division systems. The map D is not additive:

$$D(X+Y)-D(X)-D(Y)=XY(X+Y),$$

and its degree is three, whereas a nonconstant Drinfeld or Carlitz endomorphism in characteristic two is an additive polynomial of power-of-two degree. Nor is this a smooth elliptic Lattès system: infinity is a totally invariant exceptional point of the polynomial D, while a finite quotient of a separable elliptic isogeny of degree greater than one has no exceptional point, since the inverse images upstairs grow with every iterate and quotient fibres stay bounded. The relevant algebraic group is $\mathbb{G}_m$, the smooth group of a nodal cubic, not a hidden smooth elliptic division tower.

The natural characteristic-zero Dickson lift X^3-3X has the same semiconjugacy with cubing. The selected C-preimages come from roots of unity and therefore are preperiodic and have canonical height zero. The forward-orbit primitive-divisor theorem of Ingram and Silverman assumes an infinite orbit and therefore does not apply [6]. More fundamentally, these viewpoints still end at the same selected values $\Omega_{k,\ell}$; none evaluates them, and the exceptional target remains separate as stated above.

8 Boundedness and the remaining theorem

Lenstra asked whether the finite excesses are absolutely bounded [7]. Conjecture 1.1 would give the bound four, but boundedness itself is weaker than any maximal-order assertion above.

For $a\ge 0$, let

$$C_a=\mathbb{F}_{2^{2^a}}$$

be the complete finite Conway subfield at that level. For an odd prime p, put $x_p=\kappa_{f(p)}$ and $L_{p,a}=\mathbb{F}_2(x_p)C_a$.

Proposition 8.1 (finite-line criterion for absolute boundedness). *For every p and a,*

$$m_p<2^{2^a}\iff \text{there exists } c\in C_a \text{ such that } x_p+c\notin L_{p,a}^p,$$

Consequently,

$$\sup_p m_p<\infty \iff \text{for some fixed } a,\ \text{every } p \text{ has such a } c\in C_a.$$

Here $L^p=\{y^p : y\in L\}$ for a field L.

Proof. For $c\in C_a$, let $F_c=\mathbb{F}_2(x_p+c)$. Since $x_p=(x_p+c)+c$, one has $F_cC_a=L_{p,a}$. The degree $[L_{p,a}:F_c]$ divides $[C_a:\mathbb{F}_2]=2^a$, so it is prime to the odd prime p. Extension to $L_{p,a}$ is therefore injective on p-Kummer quotients: restriction followed by norm is multiplication by this invertible degree. Thus x_p+c is a p-th power in its defining field exactly when it is a p-th power in $L_{p,a}$.

The finite nimbers below 2^{2^a} are exactly C_a, which proves the first equivalence from the definition of m_p. A fixed a gives the uniform bound $2^{2^a}-1$. Conversely, choose a with 2^{2^a} larger than a given uniform bound. □

The proposition says that boundedness is uniform nonsaturation of one fixed finite affine line. It does not require any of the particular translates $0, 1, 4$ to have maximal primary order. No fixed line is presently known to work for all primes.

The universal frontier can now be stated without auxiliary machinery:

1. prove generation by the synchronized structural norm T_h, including the Conway–Fermat equalities $\delta_n = F_n$;
2. prove full primary order for the marked projective unit $y_{r,a}$ on every ordinary odd spine;
3. prove that every selected Gaussian period γ_k, equivalently every selected norm-one increment η_k, is primitive;
4. prove $\Psi_k \mid \mathrm{ord}(M_k)$ for every exceptional level.

Each item is a one-dimensional selected Kummer nonvanishing problem after the reductions above. Generic degree, trace, norm, or unmarked reciprocity data either determine a different coordinate or reconstruct the same selected coordinate. They do not supply the missing nonvanishing.

9 Formal and computational verification

The mathematical statements and the evidence are deliberately separated.

9.1 Formal boundary

The Lean 4 and Mathlib development [4, 13] proves the abstract cyclic-group power criterion, its full-primary order form, quotient-order factorization, the corrected norm identity, the exceptional lower-bound interface, and a collection of polynomial and finite arithmetic lemmas used by the reductions. It also proves the torus/Dickson semiconjugacy, the alternating two-step cubic conjugacy, and the order-four coordinate symmetry of the supersingular Conway curve. It checks the recorded factorizations of Ψ_k for $2 \le k \le 6$, the primality of the modest factors through $k = 4$, and exact order-of-two screens for the recorded factors.

The identifier `DPrimeTarget` defines

$$\forall k \ge 1, \quad \Psi_k \mid \mathrm{ord}(M_k).$$

It is not assumed and is not proved. The formal files verify ingredients of Theorem 3.1; they do not contain an end-to-end proof of Conjecture 1.1.

The formal project is built with

```
cd formal
lake build Ogdoad.Excess
```

The relevant source is `formal/Ogdoad/Excess.lean`.

9.2 Finite evidence

The finite evidence has the following exact status.

- The 126 values for odd primes $3 \le p \le 709$ are imported from OEIS A380496 and checked against the complete vendored sequence [10]. They are external tabulated data, not a proof of a universal pattern.

- The values $m_{359} = 1$, $m_{719} = 1$, and $m_{727} = 1$ have local exact finite-field certificates. The first two use marked Hilbert and crossed-norm payloads; the last uses a self-contained degree-2420 polynomial certificate. The next unsupported implementation row, $p = 733$, is an operational guardrail rather than a mathematical boundary.
- The cubic assertions C_1, C_2, C_3 are proved by exact order arguments. Levels C_4, C_5 have complete finite certificates; C_6 retains a source-assisted factorization boundary, and later checked factors are evidence only.
- The exceptional levels D'_2, D'_3 follow analytically from Proposition 7.5. Levels D'_4, D'_5 have complete finite certificates. Level D'_6 retains a source-assisted primality boundary; later partial factorizations do not certify a level.

The maintained finite-field entry points are

```
python3 experiments/ordinal_excess_probe.py
python3 experiments/cyclotomic_3k_family.py
python3 experiments/exception_column_m4.py
python3 experiments/ordinary_359_hilbert_certificate.py --full
python3 experiments/ordinary_719_crossed_certificate.py --full
python3 experiments/ordinary_727_certificate.py
```

The full 359 and 719 certificates are intentionally expensive. A successful finite run proves only its named row or level.

10 Conclusion

The finite excess problem is not presently a problem of discovering another candidate sequence. Its exact candidate is Conjecture 1.1, and Theorem 3.1 reduces it to four explicit selected-order assertions. The reductions explain both the strength of the numerical pattern and the persistence of the proof gap: all coarse finite-field invariants are compatible with more than one value of the selected Kummer coordinate.

The most economical remaining target is therefore direct. One must prove nonvanishing of the recursively selected Euler quotient in each of the four arms, with full primary valuation. Until those four statements are proved, the universal 0/1/4 rule and even absolute boundedness remain open.

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